AES T1 Exercises. Seminar 1

1. We have a loudspeaker with an effective diameter of 25 cm radiating 20mW of sound power in free field. Calculate the sound intensity level (SIL) for the loudspeaker's surface.
2. Calculate the sound power level of a source radiating 1W.
3. A speaker radiating in free field generates 80dB at a distance of 3m. Calculate the level at 5m and at 2m. Indicate what should we expect for the same situation when radiating inside a room.
4. We have a loudspeaker generating 115dB SPL at 1m with an amplifier of 1W. What will be the SPL generated at 1m if we change the power of the amplifier to 0,22W?
5. Calculate the increase of SPL experienced when two singers sing together, both generating the same level (consider not correlated sources). What will be the increment generated by a choir of N voices?
6. In a recording studio we have the air conditioning system and a fan. Together they generate a background noise of 80dB. The air conditioning without the fan generates 75dB. Calculate the SPL generated by the fan.
7. We have 2 stereo main P.A. systems installed on the beach. One system is aligned with the stage. The other one is installed at 40m from the stage to reinforce the farthest area of the audience. The left (L) and right (R) loudspeaker arrays of both systems are respectively separated 30m one from each other. The mixing console is situated forming an equilateral triangle with the L&R (PA1) system.
    a. Calculate the distance between the mixer and the stage.
    b. We turn on the left array of the first P.A. system situated on the stage (PA1-L) and it generates 118dB SPL at the mixing console's sweet spot. Calculate the total level when the L&R PA1 systems are turned on, reproducing a mono signal.
    c. Approximate the level of the complete PA1 system at a distance of 104m from the stage (centered position). Give two results, considering free field spherical and cylindrical radiation.
    d. Now we run a low frequency test with a mono sine wave of 100Hz that generates 100dB SPL in the mixer position. If we change the sonometer to measure A-weighted levels, which level should we measure?
    e. Calculate the delay needed in the second P.A. system (PA2) to align its wave fronts with the PA1 system (c=344m/s).
    f. Explain briefly what will happen if we don't insert the delay of the previous question.

8. Calculate:
    a. The level in dBV equivalent to 16V.
    b. The level in dBW equivalent to 16W
    c. The voltage equivalent to +4dBu.
    d. The voltage equivalent to -10dBV.
    e. The level in dBm equivalent to +4dBu (600Ω).

9. We generate a sine wave of 1KHz inside a mixer at its alignment level (+4dBu) and we send it to a mono master output equipped with a SMPTE RP155 peak meter.
    a. Calculate the voltage in volts that we could measure at the mono output if we have the main fader at -10dB.
    b. We use a secondary synchronized oscillator of the mixer to generate another 1KHz tone that we mixdown with the previous tone at the mono output master bus. Calculate the time delay we should introduce in this channel to cancel the sine waves.
    c. Now we plug in 2 mono tracks of a multitrack session (synth and voice) in 2 channels of the mixer, and we send them to the mono main output. With all the faders and the master fader at 0dB, the level of the synth (with the voice muted) is 19 dBu at the master's peak meter. If the level of the voice is also 19dBu (with the synth muted), will the master clip when the two channels are unmuted?
    d. Calculate in volts the total voltage measured by the meter in the previous question.

10. For the next case study, calculate the total SPL generated at the center of a square formed by four loudspeakers. (Consider both coherent and not coherent addition).
    - Each frontal loudspeaker generates 94dB SPL at the center of the square
    - Each rear loudspeaker generates 90dB SPL at the center of the square